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West Africa in Rodinia: High quality paleomagnetic pole from the ~ 860 Ma Manso dyke swarm (Ghana)

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ABSTRACT

The paleogeography of the Meso-Neoproterozoic Rodinia supercontinent remains debated partly because many stable cratons still lack reliable paleomagnetic data for this period. A new geochronological and paleomagnetic study was conducted on the NNW-trending Manso dyke swarm of southern West Africa (Ghana) to clarify the position of this unconstrained continent in Rodinia. Two U—Pb apatite ages of 857.2 ± 8.5 Ma and 855 ± 16 Ma agree with one previous baddeleyite age, indicating a ~ 860 Ma emplacement age for the Manso dykes. A characteristic remanent magnetization (ChRM) was isolated in stable single to pseudo-single domain (SD-PSD) magnetite. Well constrained site mean directions obtained for 13 dykes lead to a mean direction for the Manso dyke swarm of $D_m = 181.9°$, $I_m = -77.2°$ ($N = 13$, $\alpha_{95} = 7.6°$, $k = 30.6$), yielding a paleomagnetic pole at 177.6°E, 28.3°S, ($A_{95} = 12.7$ °K = 11.6). Two directional clusters of opposite inclination pass a reversal test (C-class) and the primary origin is supported by a positive baked contact test, satisfying all the seven R-criteria to provide the first West African Tonian key paleomagnetic pole. This key pole indicates a high latitude for the West Africa Craton during the emplacement of the ~860 Manso dykes. A compilation of reliable paleomagnetic poles for West Africa, Baltica, Amazonia and Congo-São Francisco cratons suggests that these cratons were together between ~1200 and 800 Ma in a long-lived WABAMGO configuration. We suggest that the collision of this block with Laurentia along the Grenvillian-Sunsás orogens closed the external Nuna Ocean and formed Rodinia by extroversion.

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1. Introduction

The Earth's geodynamics at the Mesoproterozoic–Neoproterozoic transition was marked by the formation of the Rodinia supercontinent, whose duration and configuration are still subject to debate (Condie, 2002; Dalziel, 1997; Evans, 2009; Hoffman, 1991; Li et al., 2013; Li et al., 2008; Meert, 2001; Meert and Torsvik, 2003; Merdith et al., 2017; Moores, 1991; Pisarevsky et al., 2003; Sears and Price, 1978; Wen et al., 2018; Wingate et al., 2002). In all models, Laurentia is considered as the central piece of Rodinia, surrounded by passive margins during the late Neoproterozoic, similar to the African plate during the Phanerozoic (Bond et al., 1984; Hoffman, 1991). Regarding the paleomagnetic database for Rodinia, Laurentia has a substantial number of reliable paleomagnetic poles between ~1270 and 1000 Ma, but the ~1000–800 Ma interval lack of high-quality data (Evans, 2009). Around Laurentia, the models place Baltica along the northeastern coast of Laurentia/Greenland in the northern hemisphere or, as an alternative, in an inverted position (Hartz and Torsvik, 2002). Siberia is often considered to be adjacent to the northern margin of Laurentia in a range of orientations: (i) with the present-day Siberia northern margin facing the northern Laurentia margin (Dalziel, 1997; Hoffman, 1991), or (ii) in

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a reverted position (Ernst et al., 2016; Rainbird et al., 1998). Paleomagnetic comparisons between Siberia and Laurentia support a relative distance between these cratons considering Siberia as a promontory of the supercontinent with smaller cratons between them (Li et al., 2008; Pisarevsky and Natapov, 2003; Pisarevsky et al., 2008). Some models have also located Siberia along the present western margin of Laurentia (Piper, 2007; Sears and Price, 1978; Sears and Price, 2000), but this position is no longer supported by paleomagnetic data (Li et al., 2008). An alternative position in the southern hemisphere near the North China Craton was also proposed (Evans, 2009). Considered as neighbors in most supercontinents (Columbia, Rodinia and Pangea), these three cratons (Laurentia-Baltica-Siberia) are defined as the "strange attractors" by Meert (2014). The Amazonia, West Africa and Rio de la Plata cratons are unconstrained by palaeomagnetic data during the ~1000–700 Ma interval, but they are usually placed in proximity to western Laurentia (Evans, 2009). Coherence between these blocks but with slight differences in orientation in different reconstructions led Meert (2014) to call them the "spiritual interlopers". Note that at the peak of Rodinia continental assembly at ~950–850 Ma, only four continents (Baltica, São Francisco, North China, and Siberia) exhibit robust paleomagnetic data (Merdith et al., 2017). This emphasizes the still scarce paleomagnetic database for Rodinia event in its "golden age".

The peak of continental Rodinia assembly precedes a drastic change in paleolatitude from a high-latitude for the Rodinian landmasses to a low-latitude paleogeography, which occurred at ~800 Ma, before its breakup at ~750 Ma (Li et al., 2008). This rotation began after the emplacement of a large superplume beneath the polar landmasses at ~840 Ma, triggering large magmatism and rifting (Li et al., 2003). Recently, some authors argued that these events can be better explained by a Tonian inertial interchange true polar wander (IITPW), but its amplitude, characteristics (single shift or TPW oscillations), and duration are debated (Jing et al., 2019; Li et al., 2004; Maloof et al., 2006; Niu et al., 2016; Swanson-Hysell et al., 2012). From an environmental point of view, no glacial deposits were observed in the landmasses during the polar position of Rodinia at ~950–850 Ma (Li et al., 2013), but more constraints are needed since climatic models depends strongly on the Tonian (1000–720 Ma) paleogeography (Donnadieu et al., 2004). The ~812–790 Ma interval is characterized by the Neoproterozoic Bitter Springs Anomaly (BSA), a large negative $\delta^{13}C$ excursion which can be associated with some indicators of increasing oxygenation of the ocean and atmosphere during the radiation of early eukaryotes (Swanson-Hysell et al., 2015b).

Therefore, a precise and reliable Tonian paleogeography is crucial to understand the impact of the Rodinia supercontinent and the magmatic events it encompasses on the Earth's system. In this contribution, we performed a detailed paleomagnetic study on the ~860 Ma Manso dyke swarm (Baratoux et al., 2019) associated with new U—Pb apatite dating to obtain the first Tonian key pole for West Africa. Our first well constrained West Africa key paleomagnetic pole allows us to propose the existence of a long-lived WABAMGO juxtaposition between West Africa, Baltica, Amazonia and Congo-São Francisco cratons, which collision with Laurentia leaded to the Apex of Rodina supercontinent.

2. Geological setting

The West African Craton is composed of two Proterozoic Shields (Reguibat Shield in the north and Leo-Man Shield in the south) stabilized at about ~2 Ga, and separated by the Upper Proterozoic–Paleozoic sedimentary Taoudeni Basin (Black et al., 1979) (Fig. 1). The Ghanaian Paleoproterozoic domain (within the Baoulé-Mossi domain) is composed of an association of granitoids and five northeastern trending greenstone belts (Bole-Nagandi, Bui, Sefwi, Ashanti, and Kibi-Winneba belts from west to east, respectively) (Feybesse et al., 2006) (Fig. 1). These greenstones and the associated sedimentary basins were deformed during the ~2100–2000 Ma Eburnean Orogeny (Bonhomme, 1962). These two Proterozoic Shields underlie Meso-Neoproterozoic Basins (Affaton et al., 1991). The older sediments of the Volta basin (Lower Voltaian Supergroup) in southern Ghana were deposited at ~1000 Ma. The southeastern limit of the Volta basin is bordered by the Dahomeyide belt, the southern extension of the Pan-African belts (Affaton et al., 1991) (Fig. 1). These Neoproterozoic events have strongly affected the northern West African Craton, where the sedimentary sequences and dykes are folded and deformed along shear zones in the Anti-Atlas Orogen (Fig. 1) (Samson et al., 2004), while the Leo-Man Shield in turn remained stable.

Twenty-six distinct dyke swarms were identified in the West African Craton by aeromagnetic mapping according to their orientation (Jessell et al., 2015). In the Leo-Man Shield, Paleoproterozoic–Mesoproterozoic dykes are represented by the ~1790 Ma Libiri swarm, the ~1790 Korsimoro swarm, and the ~1520 Essakane swarm (Baratoux et al., 2019). Paleomagnetic poles from greenstone rocks and Paleo- to Mesoproterozoic dykes were previously reported by Piper and Lomax (1973). Two Neoproterozoic dyke swarms from Ghana were recently dated by U—Pb baddeleyite at 915 ± 7 Ma for the N070° Oda dyke swarm, and 867 ± 16 Ma for the N355° Manso dyke swarm (Baratoux et al., 2019) (Fig. 1). A younger generation of basaltic dykes was also recognized in the northeastern part of Ghana/southern Burkina Faso with an U—Pb baddeleyite age of 198 ± 16 Ma (Hounde dyke swarm) and was related to the Central Atlantic Magmatic Province (CAMP) (Baratoux et al., 2019).

In this study, we sampled 15 dykes of the ~860 Ma NNW-trending Manso dykes and one dyke (GH07) of the ~915 ENE- trending Oda dyke swarm (Fig. 1). The Manso dykes have mainly a NNW-direction conjugated to a NNE-direction for some branches of the dyke swarm and are ~50–100 m wide (geophysical signature), but their contact with the host rock was not observed in the field. They are fresh, coarse to medium-grained dolerites composed mainly of plagioclase and clinopyroxene (augite) with Fe—Ti oxides. Rare orthopyroxene, altered olivine, baddeleyite, and sulfide were also observed (Baratoux et al., 2019). Fine-grained lamprophyre dykes containing micas and some titanite occur in the Ahafo mine. These Neoproterozoic dykes are undeformed, and they crosscut the Paleoproterozoic basement and Paleoproterozoic regional tectonic structures.

3. Methods

3.1. Sampling

Due to the dense vegetation and the occurrence of a thick (up to ~30 m) laterite cover, the outcrops of mafic dykes in Ghana are restricted to the rivers, isolated blocks (Fig. 2A, B), and fresh outcrops from mine pits. In March 2019 we sampled 121 oriented cylindrical cores using a portable gasoline-powered rock drill, as well as eight hand-samples in the Ahafo mine pit (Newmont Company), for a total of 16 sites (or dykes). According to the field orientation, 15 dykes are from the ~860 Ma NNW-trending Manso dyke swarm and one dyke (GH07) is from the ~915 Ma E-trending Oda dyke swarm (Antonio et al., 2019; Baratoux et al., 2019). The number of samples by site (5–8) were restricted essentially due to agreement with local communities and the mining company. Both cores and hand-samples were oriented using solar and magnetic compasses, and no declination difference was observed. The 16 sites sampled cover a vast geographical area (~41,500 km^2) from the capital Accra in southeast Ghana to Sunyani in the northwest *via* Cape Coast in the south (Fig. 1 & site coordinates provided in Table 1).

3.2. Geochronology

Two samples (GH05 and GH16) were selected for U—Pb geochronology on apatite. Apatite separation was performed in the Géosciences Environnement Toulouse (GET, France) laboratory. Hand-samples were crushed and sieved to collect the mineral fraction below 400 μm. The

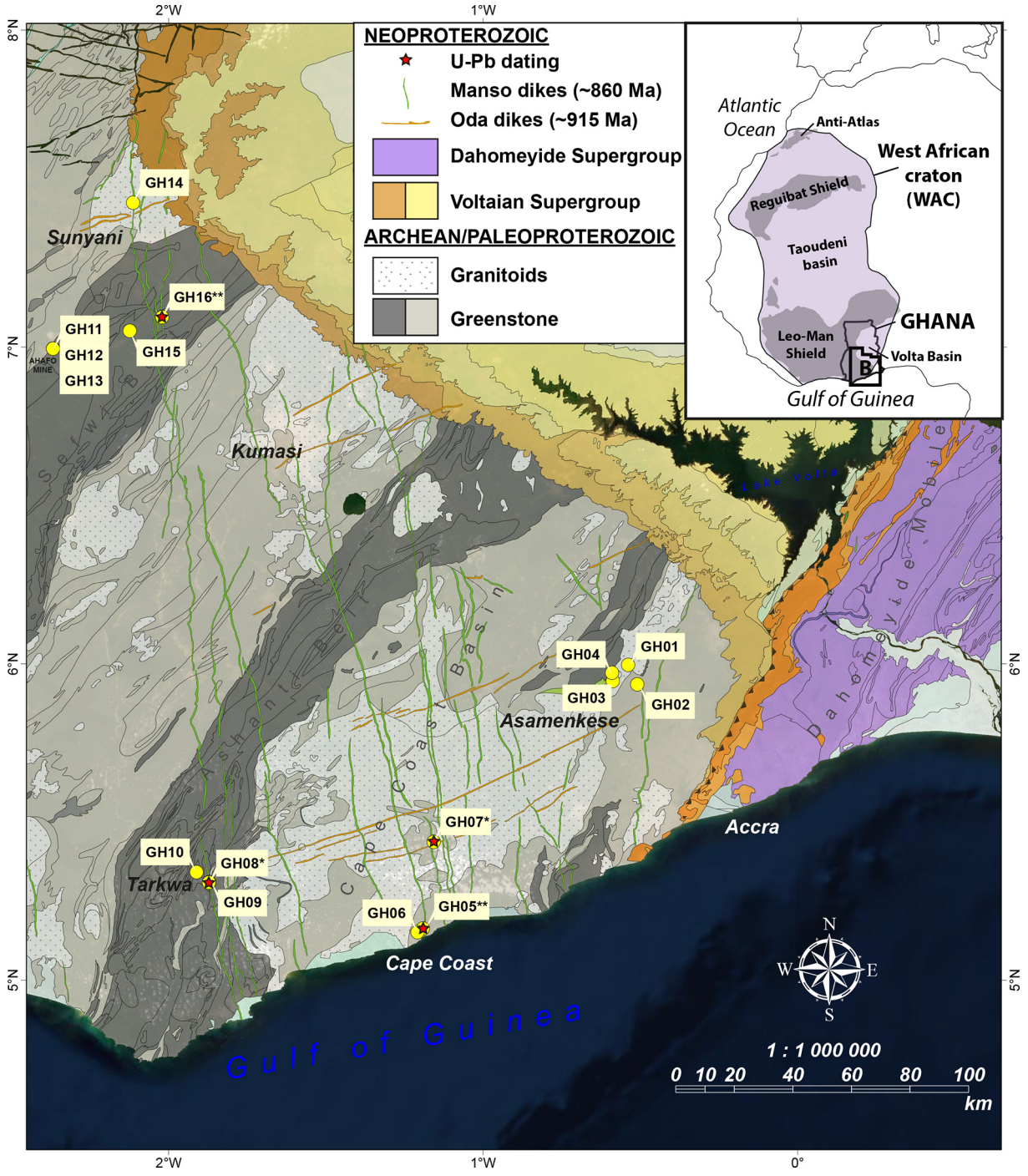


Fig. 1. Inset: Location of the study area in the West African Craton (WAC). Geological map of the Neoproterozoic units of Ghana with sampling location for geochronology (red stars) and paleomagnetism. *: U—Pb baddeleyite age of GH08 = 867 ± 16 Ma (Manso) and GH07 = 915 ± 7 Ma (Oda) from Baratoux et al. (2019). **: U—Pb apatite ages of GH05 = 857.2 ± 8.5 Ma and GH16 = 855 ± 16 Ma from this study. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

low-density minerals and clay fraction were removed using a Wilfley table. Heavy minerals were then isolated using heavy liquids (tetrabromoethane and diiodomethane, with respective densities of 2.967 and 3.325 g/cm^3). Magnetic minerals were consequently removed with a Franz magnetic separator. Finally, the apatite grains were handpicked using a binocular microscope before being mounted into epoxy puck and polished. U—Pb data were acquired at the Isotopic Geochemistry Laboratory in Federal University of Ouro Preto (UFOP, Brazil) using a 193 nm HelEX Photon Machine coupled with a ThermoScientific Neptune Plus Multicollector (LA-MC-ICP-MS). A beam spot size of 85 μm was used with beam energy densities of 6 J/cm^{-2}, and a 6-Hz repetition rate. During apatite U—Pb measurement sequences, the 91,500 zircon (Wiedenbeck et al., 1995) was used as a primary reference standard, while the Durango apatite (McDowell et al., 2005), the 401 apatite (Thompson et al., 2016), and the Madagascar apatite (Thomson et al., 2012) were used as secondary reference material in order to correct for matrix match effects as well as to constrain and verify the corrections accuracy and reproducibility. Data reduction and correction was carried out with the SATURN package of the laboratory of Ouro Preto. Apatite U—Pb ages of the studied

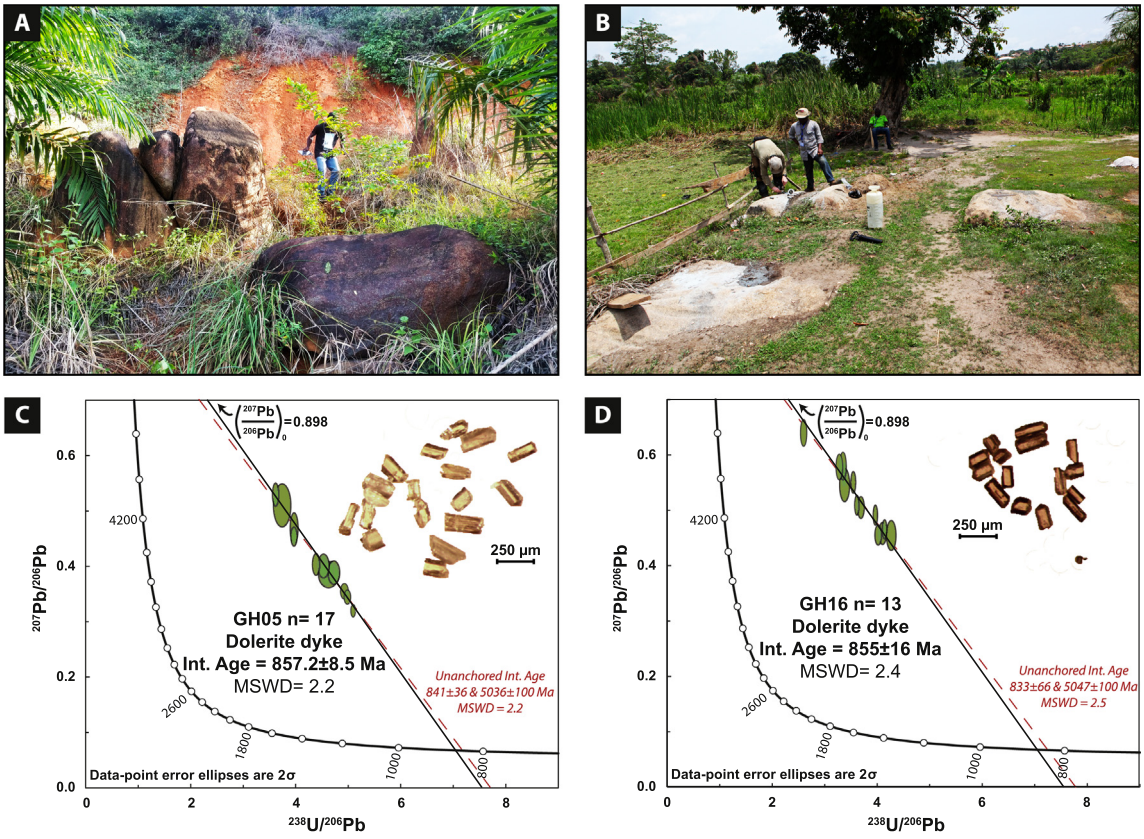


Fig. 2. A and B: Field photographs of the Manso dolerite dykes at GH05 and GH16 sites, respectively. C and D: U—Pb Terra-Wasserburg diagrams for apatite dating of the Manso dolerite dykes at GH05 (C) and GH16 (D) sites. The upper intercept is anchored to the initial Pb Pb ratio (0.898) that was calculated based on Pb model of Stacey and Kramers (1975) for an emplacement age of ~867 Ma (Baratoux et al., 2019). Unanchored ages are also illustrated (in red). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

samples are reported as isochron ages calculated as lower-intercepts on a Terra-Wasserburg Concordia diagram using the Isoplot 4.15 software (Fig. 2) (Ludwig, 2009). The grains analyzed from each sample are igneous co-genetic apatites. Initial common Pb value was anchored to a $^{207}Pb/^{206}Pb$ value of 0.898 (Stacey and Kramers, 1975) according to the U—Pb baddeleyite (ID-TIMS) crystallization age of 867 ± 16 Ma (Baratoux et al., 2019). All the apatite isotopic data, at 2σ level, are reported in the supplementary material 1. Additional information on the analytical conditions, and the Terra-Wasserburg Concordia diagrams of reference materials, are provided in the supplementary material 2.

3.3. Paleomagnetism

Oriented blocks were drilled in Geosciences Montpellier (France). Preparation of standard specimens (2.2 cm height) and alternating field (AF) demagnetization for a pilot study were carried out at the GET (Toulouse, France) using a JR5-A spinner magnetometer and a LDA-3 AF demagnetizer (AGICO) in a MMLFC shielded room design to reduce the effect of ambient magnetic field (<200 nT). For the remaining samples, the characteristic remanent magnetization (ChRM) was isolated by stepwise AF and thermal demagnetization performed in a magnetically shielded room with ambient field <500 nT at the Laboratorio de Paleomagnetismo of the University of São Paulo (USPMag, Brazil). An AF pre-treatment until 10 mT was performed before thermal demagnetization, to eliminate an eventual low-coercivity, viscous component. AF and thermal demagnetizations were performed using a vertical 2G-Enterprises™ DC-SQUID magnetometer with a RAPID automatic sample changer (Kirschvink et al., 2008). Stepwise thermal demagnetization of 50 °C (until 500 °C) and 20 °C (until 700 °C) were carried out using a TD-48 furnace device. Only principal component analysis (PCA) (Kirschvink, 1980) was used to determine the remanence directions using orthogonal vector diagrams (Zijderveld, 1967). Site mean directions and paleomagnetic poles were calculated by Fisher's (1953) statistics using the PALEOMAC package (Cogné, 2003). Paleogeographic reconstructions were performed using the GPlates software (Müller et al., 2018). Magnetic mineralogy was investigated under optical microscopy and using a Scanning Electron Microscopy (SEM JEOL JSM 7100F TTLS LV – EDS/EBSD) at the Centre de Micro Caractérisation Raimond Castaing (Toulouse, France). High-temperature thermomagnetic curves (susceptibility *versus* temperature) were conducted at the Toulouse, France) in an argon atmosphere using a CS-3 apparatus coupled to the KLY-3 Kappabridge (AGICO, Brno, Czech. Republic). In addition, hysteresis loops, isothermal remanent magnetization (IRM) and first-order reversal curve (FORC) for selected samples were performed at the LABGeo, Instituto Oceanográfico, University of São Paulo (Brazil) using a MicroMag-VSM, Model 3900 (Princeton Measurements Corporation). FORC diagrams were processed using the Forcot software (Berndt and Chang, 2019).

4. Results

4.1. Geochronology

The GH05 site is situated near Cape Coast, between the towns of Yamoransa and Biriwa (southern Ghana) (Fig. 1). It is a 20-50 m wide dyke disintegrated into several blocks of coarse-grained relatively fresh dolerite, with ~0.5 cm of weathering crust (Fig. 2A). The second dated site GH16 is located within the Bechem city (SE of Sunyani) (Fig. 1). Multiple blocks of metric size (*in situ*) were observed in an

Table 1
S.lat, S.lon are the site latitude and longitude. n/N - number of specimens used in mean directions/number of analyzed specimens; Dec – Declination; Inc. – Inclination; R – the resultant vector, α95 (*A95*) is the radius of the 95% cone of confidence and k (*K*) is the precision parameter - Fisher's statistical parameters (Fisher, 1953) for the mean directions (*mean virtual geomagnetic poles – VGPs*); P.Lat – pole latitude; P.Long – pole longitude. Values of mean sites directions are indicated in bold for the positive, negative, and combined polarities. *: U—Pb baddeleyite age of GH08 = 867 ± 16 Ma (Manso) and GH07 = 915 ± 7 Ma (Oda) from Baratoux et al. (2019). **: U—Pb apatite ages of GH05 = 857.2 ± 8.5 Ma and GH16 = 855 ± 16 Ma from this study. ¥: Site with an attempted baked contact test (BCT). γ (γc = critical) is the angle calculated between the mean directions of reversed and normal polarities (McFadden and McElhinny, 1990).

Sites (Samples)	S.lat (°N)	S.lon (°E)	Lithology	Characteristic remanent magnetization (ChRM)						VGP	
				n/N	Dec (°)	Inc (°)	R	k	α95 (°)	P. Lat (°N)	P. Long (°E)
Cluster (A+) with downward inclination (positive)											
GH03 (A-O)	5.946	359.413	Dolerite dyke, medium to coarse grained	30/40	22.6	51.8	29.07	31	4.8	55.3	33.8
GH11 (OB1-OB2)	6.994	357.633	Lamprophyre dyke	13/13	320.6	86.7	12.91	314.8	3.6	11.1	352.7
GH12 (OB3-OB4)¥	6.994	357.633	Lamprophyre dyke	14/17	87.9	86.8	13.87	100.2	4	6.2	3.4
GH13 (OB5)	6.994	357.633	Lamprophyre dyke	14/14	82.1	88.2	13.98	548	1.7	6.5	0.6
GH15 (A-H)	7.051	357.876	Dolerite dyke, fine to medium grained	6/8	3.7	63.9	5.93	77.2	7.7	51.3	1.2
Mean positive polarity				**5 sites**	**16.6**	**76.7**	**4.81**	**21**	**17.1**	**26.4**	**3.9**
							***R* = 4.54**	**K = 8.7**	**A95 = 27.5**		
Cluster (A-) with upward inclination (negative)											
GH01 (A-L)	5.996	359.462	Dolerite dyke, medium grained	9/24	207.6	−57.2	8.6	20	11.8	−49.5	213.3
GH02 (A-H)	5.935	359.490	Dolerite dyke, medium grained	11/28	220.2	−70.9	10.72	35.6	7.8	−31.2	205
GH05 (A-H)	5.163	358.809	Dolerite dyke, medium grained**	10/15	137.6	−85.2	9.84	57.8	6.4	−11.9	171.5
GH08 (A-H)	5.309	358.130	Dolerite dyke, medium grained*	17/17	136.9	−78.2	16.99	1134	2.4	−21.2	161.5
GH09 (A-F)	5.307	358.129	Dolerite dyke, medium grained	9/12	142.2	−72.3	8.89	67.2	6.3	−29.8	155.7
GH10 (A-E)	5.341	358.088	Dolerite dyke, fine grained	8/19	127.8	−68.8	7.81	37.4	9.2	−26.3	145.3
GH14 (A-F)	7.454	357.887	Dolerite dyke, medium grained	12/15	152.7	−80.4	11.97	338.6	2.4	−23.5	167.8
GH16 (A-K)	7.095	357.979	Dolerite dyke, coarse grained**	12/22	198.1	−78.6	11.94	168.4	3.4	−27.7	184.5
Mean negative polarity				**8 sites**	**172.8**	**−77**	**7.81**	**37.1**	**9.2**	**−29.2**	**173.7**
							***R* = 7.47**	**K = 13.2**	**A95 = 15.8**		
MEAN MANSO POLE - COMBINED (by sites)				**13 sites**	**181.9**	**−77.2**	**12.6**	**30.6**	**7.6**	**−28.3**	**177.6**
							***R* = 11.97**	**K = 11.6**	**A95 = 12.7**		
Angles of reversal test								**γ = 5.4°**	**γc = 16.2°**		
Mean combined (by specimens)				165	187.9	−75.9	156.6	19.6	2.5		
Mean secondary (by site)				10 sites	357.7	1.3	9.23	11.7	14.7	84.3	197.8
Sites not considered in the Manso pole.											
GH04 (A-U)	5.971	359.409	Dolerite dyke, medium grained (not *in situ*)	34/35	114.6	13.5	33.12	37.7	4.1	−23.5	79.3
GH06 (A-E)	5.150	358.792	Dolerite dyke, fine grained (not *in situ*)	8/12	330	23.8	7.65	20	12.7	59.4	284.1
GH07 (A-H)	5.436	358.843	Dolerite dyke, fine grained (~915 Ma)*	28/29	245.3	−62.7	26.99	26.7	5.4	−21.1	222.3
Baked contact test											
Mean baked - BCT (GH12)	6.994	357.633	Foliated granodiorite	9/11	117.9	80.3	8.93	112.8	4.9	−2.9	13.6
Mean unbaked -BCT (GH12)	6.994	357.633	Foliated granodiorite	9/14	291.3	31	8.83	46.7	7.6	22.1	282.7

open area (Fig. 2B). The dyke is a coarse-grained dolerite showing a fresh doleritic texture (*i.e.* intergranular subophitic).

Apatites from the doleritic samples (GH05 and GH16) exhibit a nearly perfect euhedral prism shape (Fig. 2C, D) with grain size ranging between 90 and 300 μm. The isotopic data obtained for the GH05 dolerite sample displays variable proportions of common Pb with $^{207}Pb/^{206}Pb$ values varying between 0.32 and 0.53. The anchored lower intercept age for this sample is 857.2 ± 8.5 Ma with a MSWD of 2.2 using 17 apatites (Fig. 2C). Sample GH16 (dolerite dyke) displays variable proportions of common Pb with $^{207}Pb/^{206}Pb$ values between 0.45 and 0.64. Data for this sample define an anchored lower intercept age of 855 ± 16 Ma with a MSWD of 2.4 using 13 apatites (Fig. 2D). The unanchored plots give identical ages for GH05 and GH16 respectively within the error of 841 ± 31 Ma (MSWD = 2.2) and 833 ± 66 Ma (MSWD = 2.5), respectively, but the anchored ages were preferred following Chew et al. (2011).

4.2. Paleomagnetic results

Natural remanent magnetization (NRM) values for the dolerite dykes vary between ~0.1 and 9.8 $A.m^{-1}$. Samples showing scattered NRM directions and higher intensity values of 270–555 $A.m^{-1}$ (*e.g.* GH07A, B) were discarded, most probably due to lightning effects.

A stable magnetic component was reached for the NNW-trending Manso dolerite dykes after removing a secondary/viscous component. Linear behavior is generally observed with two well-defined segments in the Zijderveld plots with a "high coercivity/temperature component" revealed above AF values of 17 mT (Fig. 3A, E, and F) and temperature values range of ~540–580 °C (Fig. 3B, C, and D). These unblocking temperatures (T_{ub}) point toward magnetite as the main carrier of the high-temperature component. Calculated by PCA analysis, a characteristic remanent magnetization (ChRM) is revealed for a first cluster (A+) of five sites showing a northern direction with a positive inclination (Fig. 3A, B). A second cluster (A-) is composed of eight sites with a ChRM of southern direction with a steep negative inclination (Fig. 3C, D, and E).

Most low coercivity-temperature components yield northern directions with shallow inclinations with the subsequent secondary site-mean direction is D_m = 357.7°, I_m = 1.3° (α95 = 14.7°, k = 11.7), close to the present dipolar field (PDF) (Fig. 4A). Three sites (GH03, GH10, GH14) show a different secondary direction of north-western directions.

A total of 165 specimens (13 sites) was used to calculate the Manso pole using the high coercivity-temperature components. Using the sites with positive inclinations (D_m = 16.6°, I_m = 76.7°, n = 5, k = 21) and the sites with negative inclinations (D_m = 172.8°, I_m = −77°, n = 8, k = 37.1), the Manso dyke swarm passes a reversal test with a critical

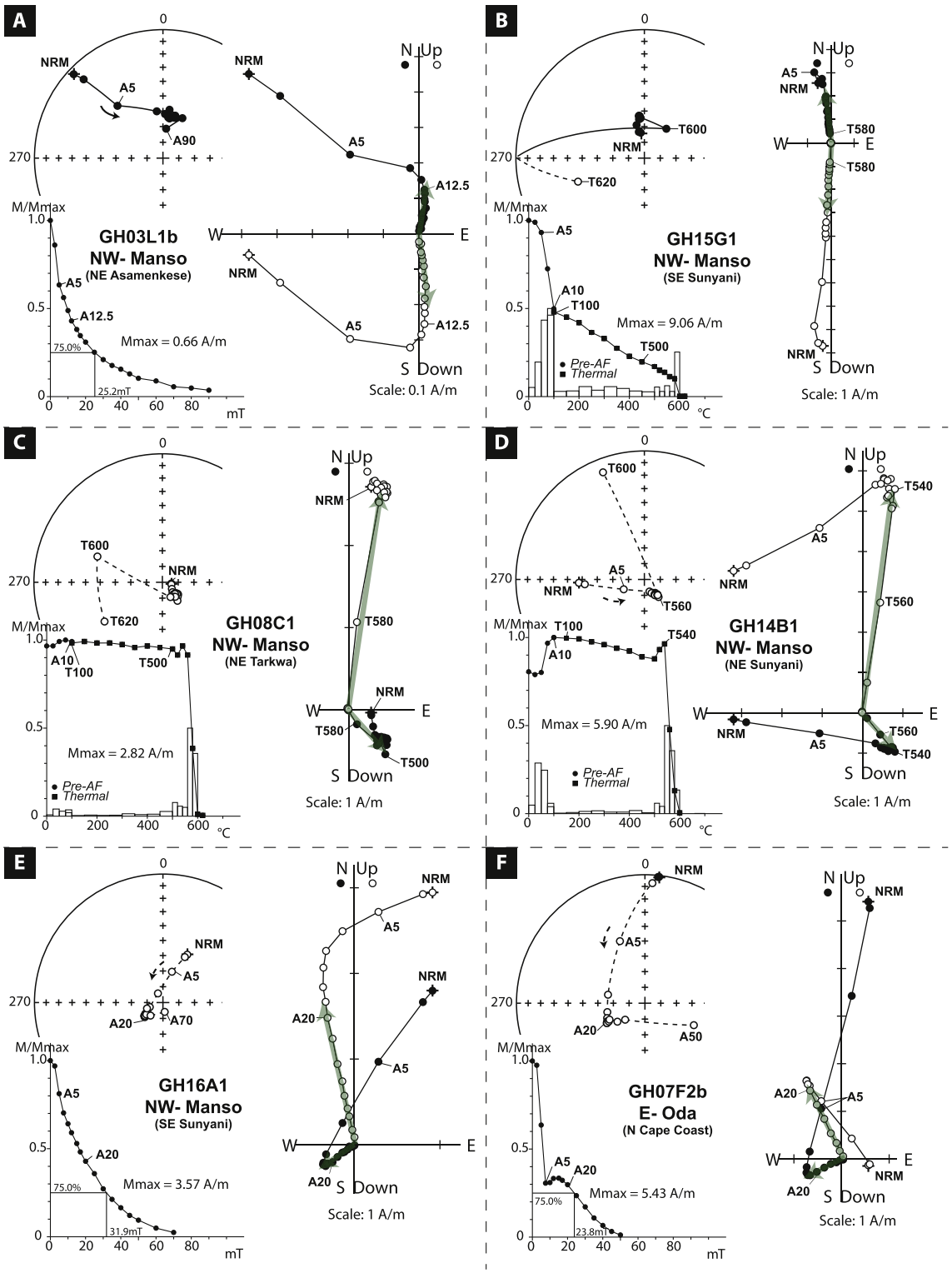


Fig. 3. Representative demagnetization plots of Neoproterozoic dykes for different geographic sites in Ghana after AF and thermal demagnetizations. Equal-area stereonets (filled (open) symbols represent positive inclination), Zijderveld plots (vertical/horizontal projections shown by open/filled circles), and Magnetization intensity decay curves (M/Mmax) are indicated for each example. Values of 75% of the NRM decay (in mT) and unblocking temperature spectra are indicated for the AF and thermal demagnetization curves, respectively. NRM = natural remanent magnetization. Numbers, T100 (A5), indicate thermal (AF) demagnetization step.

gamma of 16.2° and a difference of 5.4° between normal and reverse axes (reversal test of C-class) (McFadden and McElhinny, 1990). The Manso pole (13 sites) was calculated using the 5 sites of positive inclinations and the 8 sites of negative inclinations (Fig. 4C) giving a site mean direction of $D_m = 181.9°$, $I_m = -77.2°$ ($\alpha 95 = 7.6°$, $k = 30.6$), yielding to a paleomagnetic pole located at 28.3°S and 177.6°E (A95 = 12.7 °K = 11.6) (Table 1; Fig. 4C, D).

An iterative cutoff of 37.6° was determined using the 13 sites with no site exclusion (Vandamme, 1994), yielding a VGP dispersion (S-value) of 23.8° (Fig. 4D). High dispersion is expected for results with high

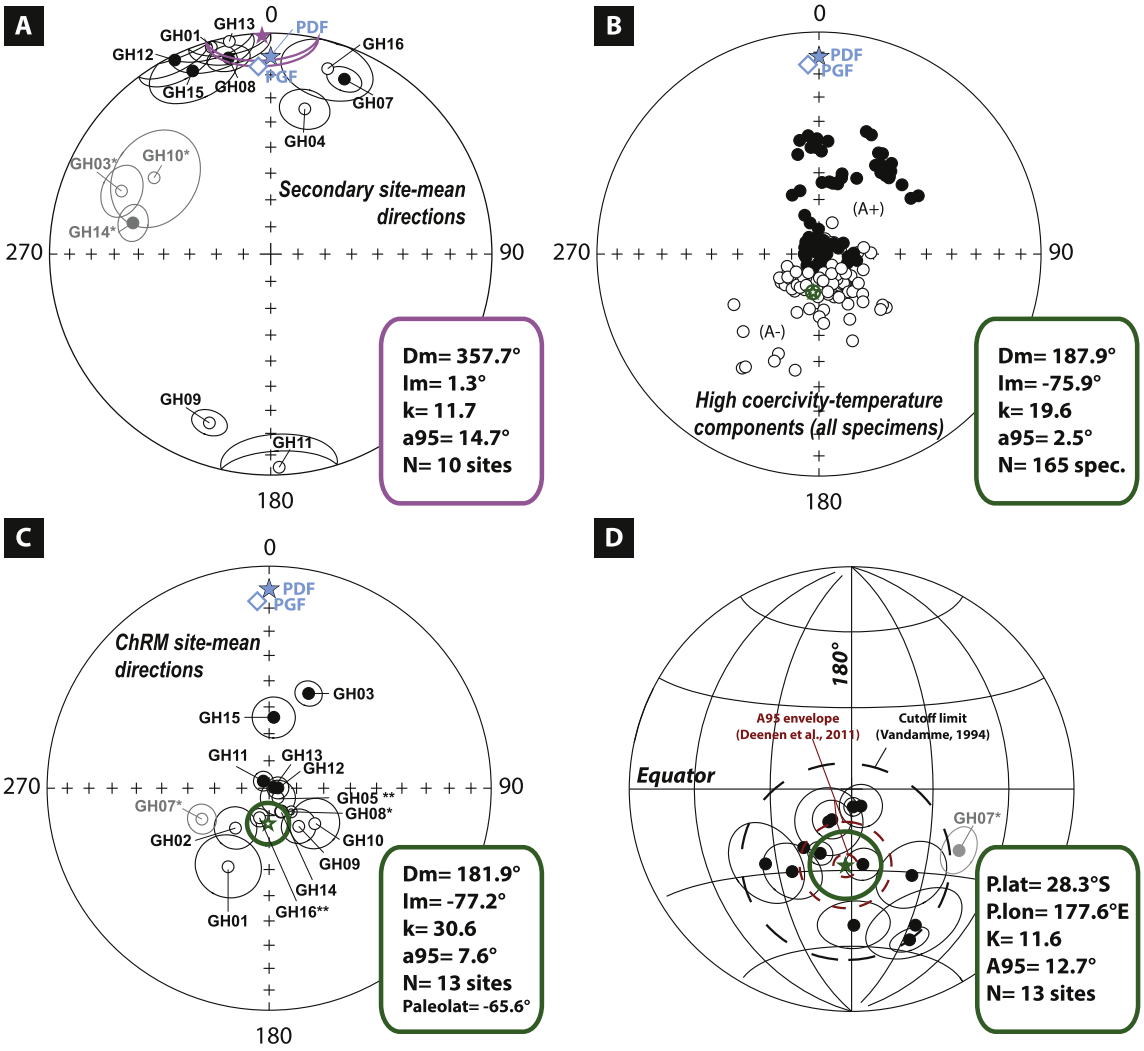


Fig. 4. Sample and site mean directions for the Manso dykes. A: Secondary site-mean directions calculated from the low coercivity/temperature components (*: not considered in the mean secondary directions). B: Specimen mean directions of the high coercivity/temperature components (in green). C: Site mean directions for the Manso dykes (in green). Sites/specimens with positive and negative inclinations are represented with filled (open) symbols represent downward (upward) inclinations. Confidence circle (α95) around the means are indicated. PDF – Present dipolar geomagnetic field; PGF – Present geomagnetic field. D: Dispersion of site mean virtual geomagnetic poles (VGPs) and paleomagnetic pole calculated for the Manso dykes (Table 1) represented in a Schmidt projection. Deenen et al. (2011)'s A95 envelope (4.3°–16.3°) and the iterative cutoff limit (37.6°) of Vandamme (1994) are indicated. *: U–Pb baddeleyite age of GH08 = 867 ± 16 Ma (Manso) and GH07 = 915 ± 7 Ma (Oda) from Baratoux et al. (2019). **: U–Pb apatite ages of GH05 = 857.2 ± 8.5 Ma and GH16 = 855 ± 16 Ma from this study. Direction and VGP for the ~915 Ma site (GH07) are indicated in grey, not considered for the Manso pole. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

inclination values as suggested by paleosecular models of latitudinal dependence of S (*e.g.* Model-G from McFadden et al. (1988)). The A95 for the Manso pole is equal to 12.7° which is comprised within the A95 envelope (4.3°–16.3°) of Deenen et al. (2011) (see Fig. 4D). Altogether, these characteristics suggest that our sampling of the Manso dolerite dykes average the paleosecular variation.

Though based on a single site (GH07), a stable ChRM of steep negative inclination and southwestern direction was obtained for the ~915 Ma *E*-trending Oda dyke, similar to the A-cluster of the Manso dykes (Fig. 3F) providing a VGP at 21°S and 222.3°E.

4.3. Baked contact test

Due to the weathering cover, host rocks could only be sampled in the Ahafo mining pit (Newmont Company) in the Sefwi Belt (Fig. 5), to attempt a baked contact test (BCT). Fig. 5A shows the sampled outcrop where two undeformed lamprophyre dykes (GH11 as illustrated in Fig. 5B, and GH12) are cutting the granodiorite host rock deformed during the ~2000 Ma Eburnean orogeny (Feybesse et al., 2006). Two oriented blocks (OB3 and OB4) were collected for the GH12 dyke (Fig. 5A). To attest the primary origin of the magnetization carried by the dyke GH12, we sampled one oriented block (OB6) within the baked zone into a two- branched dyke. The oriented blocks OB7 and OB8 were sampled at ~1 m and ~ 30 m respectively from the GH12 dyke. Thermal demagnetizations revealed a high unblocking temperature interval (T_{ub}) of 520–620 °C and high stability for the ChRM of the baked host rock at the contact (OB6) with a site-mean of $D_m = 117.9°$ and $I_m = 80.3°$ ($\alpha95 = 4.9°$, $k = 112.8$) (Table 1; Fig. 5C, D). The host rock at the contact shows the same ChRM direction of the GH12 dyke ($D_m = 87.9°$ $I_m = 86.8°$, $\alpha95 = 4°$, $k = 100.2$) (Table 1; Fig. 5C, D). The secondary components are close to the present field with a northwestern direction and a low inclination for the dyke and the host rock. At ~1 m from the dyke, the OB7 block shows a significantly different behavior where thermal and AF demagnetizations reveal a more unstable magnetization (Fig. 5C), nevertheless magnetic vectors were defined for this block with two specimens providing a direction close to the dyke's direction

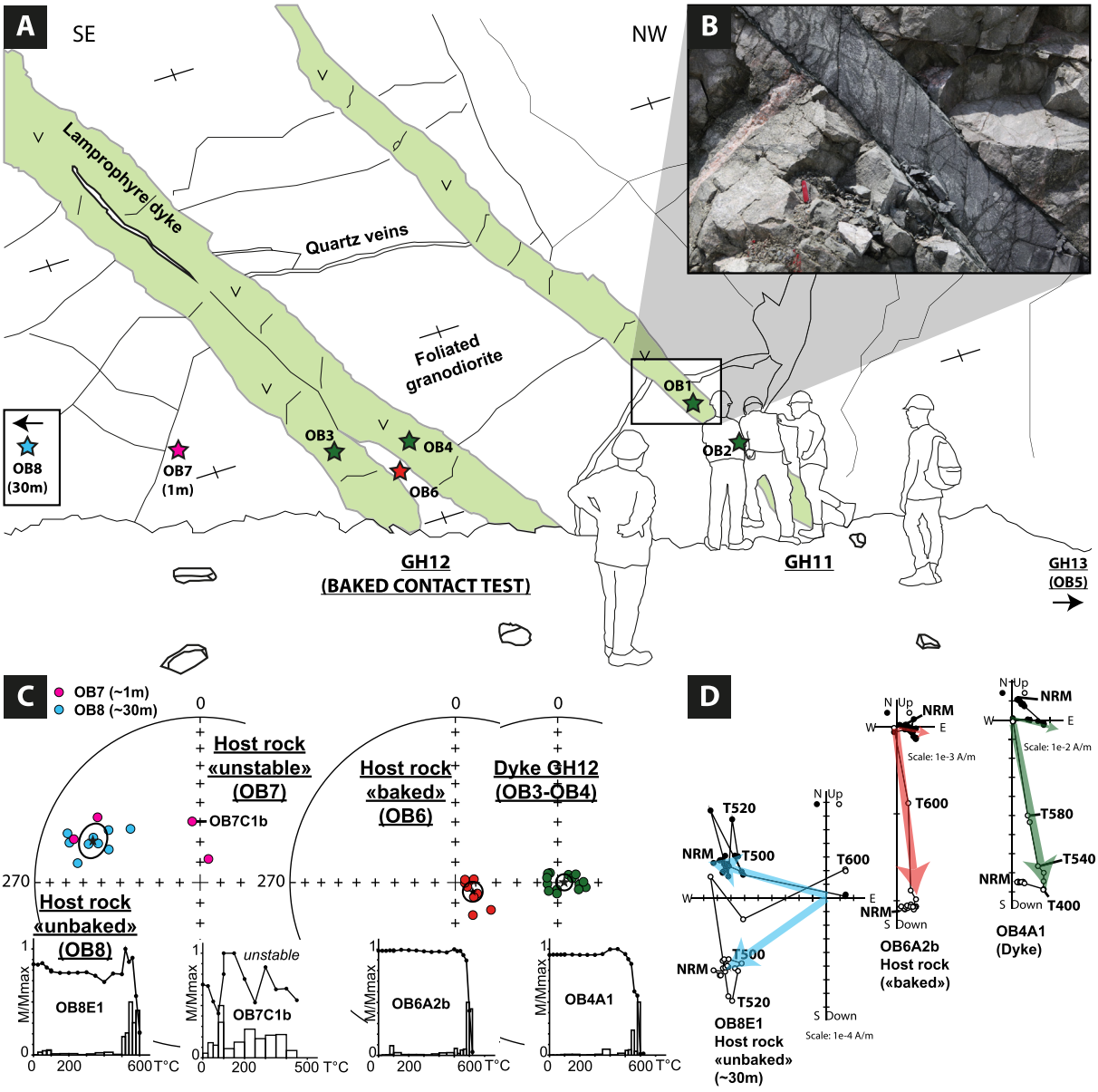


Fig. 5. Baked contact test for the GH12 dyke (~1 m in width). A: Geological sketch of the outcrop with location of the sampled oriented blocks (star) of dykes (in green), and the host rocks at contact (in red), at ~1 m (in pink), and at ~30 m (in blue) from the contact. B: Field photograph of the GH11 lamprophyre dyke, 0.30 m wide, located in (A). C: Equal-area stereonets (filled (open) symbols represent positive inclination) of site-mean directions for the GH12 dyke (in green) and the subsequent host rocks at contact (in red), at ~1 m (in pink), and ~ 30 m (in blue) from the contact. Magnetization intensity decay curves (M/Mmax) are indicated. D: Zijderveld plots for representative specimens after thermal demagnetization from the GH12 dyke (OB4A1), the host rock at contact (OB6A2b) and at ~30 m (OB8E1). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

and two specimens showing a western direction with low positive inclination (Fig. 5C). This western direction was also disclosed for the oriented block collected at ~30 m from the contact (OB8). Differently from the OB7 block, specimens from the OB8 block are well-clustered (D_m = 291.3° I_m = 31°, α95 = 7.6°, k = 46.7) with a stable ChRM until ~500 °C. This baked contact test can be considered as positive and attests to the primary origin of the stable ChRM disclosed in the Manso dykes.

4.4. Magnetic mineralogy

Most opaque phases show composite texture of ilmenite-magnetite and probably titanomagnetite (Haggerty, 1991). Fig. 6A shows a representative magnetite with ilmenite lamellae in a coarse-grained dolerite dyke (GH16). Intergrowth textures with ilmenite exsolutions are generally related to a stable thermoremanent magnetization (TRM) which supports our interpretation of the paleomagnetic results (Evans and Wayman, 1974). Fig. 6B shows three thermomagnetic curves, two for the Manso (GH08 and GH14) and one for the Oda dyke (GH07). The sample GH08A shows a curve with a reversible behavior between the heating and cooling steps whereas the cooling curve for samples GH14B3 and GH07B3 are not perfectly reversible. All samples show a Curie temperature (T_c) at about 560–580 °C, and a Hopkinson peak for GH08 and GH14 just below T_c (Dunlop, 1974), characteristic of fine-grained pure magnetite. The Day plot (Fig. 6C) and FORC diagrams (Fig. 6D) indicate domain states mainly in the stable single-domain (SD) to pseudo-single domain (PSD) fields. Values of Mrs./Ms. higher than 0.10, and the strong proportion of 60–40% of single domains magnetite in the Day plot are consistent with the narrow unblocking temperatures above 540 °C. FORC diagram of sample GH08H1 is typical of

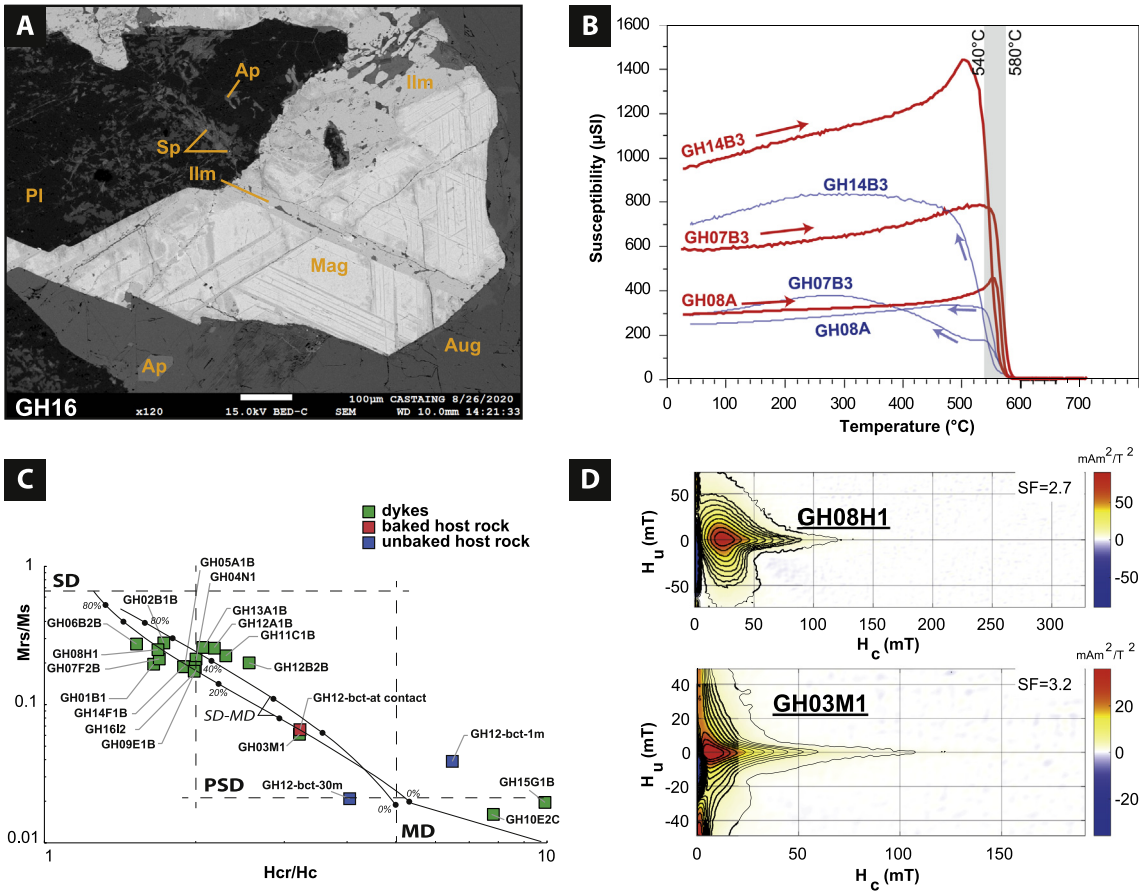


Fig. 6. A: SEM-BSE micrograph of titanomagnetite (Mag) with exsolutions of ilmenite (Ilm), augite (Aug), plagioclase (Pl), spharelite (Sp), and apatite (Ap) for the GH16 dyke. B: Heating (in red) and cooling (in blue) thermomagnetic curves for three samples (GH14B3, GH07B3, and GH08A). C: Day plot for the Manso dykes and the host rocks with SP-MD mixture lines indicating the single domain proportions. SD: single domain, PSD (or vortex state): pseudo-single domain, MD: multidomain. D: First-order reversal curve (FORC) diagrams for the GH08H1 and the GH03M1 specimens. SF is the smoothing factor used by the FORCOT software. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

the PSD behavior (Roberts et al., 2014), with SD-like magnetite dominance of the magnetic assemblage as shown by the peak value of ~25 mT with closed contours on the $H_u = 0$ axis (Fig. 6D). Conversely, the GH03M1 sample show a strong proportion of multi-domain (MD) grains mixed with SD grains as suggested by a large coercivity distribution on the H_u axis. This SD/MD behavior is also confirmed by the position of GH03M1 in the Dunlop (2002) mixing curves of the Day plot (Fig. 6D). GH15 sample fall in the MD field of the Day diagram which suggests a strong proportion of MD-like magnetite in the magnetic assemblage, consistent with the strong decay of ~50% of its NRM at 10 mT during AF pre-treatment (Fig. 3B). The host rocks fall into the PSD domain for samples at the contact, whereas samples far from the contact fall into the MD field of the Day plot (Fig. 6C), suggesting a less stable behavior for these rocks. This further confirms that the host rock at the contact was mineralogically affected and re-heated during the dyke's intrusion.

5. Discussion

5.1. Reliability of the Manso paleomagnetic pole (R-criteria)

The Manso paleomagnetic pole was calculated with 13 sites of NNW- trending mafic dykes in Ghana and satisfies all seven criteria of the revised "R" reliability index (Meert et al., 2020). (R1) The Manso dyke swarm is well-dated by multi-method radiometric dating with one U—Pb baddeleyite (ID-TIMS) age of 867 ± 16 Ma ($^{207}Pb/^{206}Pb$ weighted mean age) (GH08) (Baratoux et al., 2019) and two identical (within error) U—Pb apatite ages of 857.2 ± 8.5 Ma (GH05) and 855 ± 16 Ma (GH16) (Fig. 2). Given that the U—Pb apatite system (Chew and Spikings, 2015) has a lower closure temperature (550–350 °C) than the U—Pb baddeleyite system (700–1100 °C) (Heaman and LeCheminant, 2001), these identical ages within error suggest the U—Pb apatite system recorded the age of crystallization. Using the U—Pb baddeleyite $^{207}Pb/^{206}Pb$ weighted mean age of 867 ± 16 Ma (*or the Concordia age: 855 ± 10 Ma*) (Baratoux et al., 2019) combined with the two new U—Pb apatite ages, a mean age of 858.6 ± 6.7 (*856.1 ± 6*) Ma can be calculated for the emplacement of the NNW-trending Manso dyke swarm. (R2) ChRM directions were isolated on 13 sites (165 specimens) by thermal and AF demagnetizations and no difference was observed between the two demagnetization methods. Moreover, all vectors were well-defined by stable linear segments reaching the origin (Fig. 3), and analyzed by PCA analysis using Zijderveld plots and equal-area projections (Kirschvink, 1978; Zijderveld, 1967). Our new Manso pole is characterized by a value of precision K of 11.6, above the lower limit of 10 proposed by (Meert et al., 2020). In addition, the A95 of 12.7° is comprised within the Deenen et al. (2011)'s interval (4.3°–16.3°) showing that the pole averages the paleosecular variation (Fig. 4D). No VGPs were excluded by the iterative cutoff of Vandamme (1994). (R3) The magnetic properties of these mafic dykes confirm their remanent magnetization is carried by a magnetic assemblage dominated by SD-like magnetite (Fig. 6). (R4) A positive baked contact test for the GH12 dyke confirms the remanence

is primary. In addition, a primary origin is also suggested by the narrow and high unblocking temperatures (T_{ub}) (540–580 °C) of the Manso dykes which are above those of the undisturbed U—Pb apatite system (550–350 °C). (R5) These undeformed dykes were sampled in a vast geographic area in Ghana considered as stable and without evidence of deformation related to the Pan-African belts or younger tectonic events. Moreover, geochronological and paleomagnetic results of distant sites (*e.g.* between GH05 and GH16, which are 233 km apart from each other) support a strong regional consistency in our results. The younger events in the area were associated with the magmatic activity of the Central Atlantic Magmatic Province (CAMP) and the Atlantic rifting between Africa and South America (Baratoux et al., 2019). These events produced a low-temperature regional influence of <120 °C (Fernie et al., 2018), not sufficient to affect the primary magnetization of the dykes. This could be consistent with the secondary components of northwestern directions with shallow inclinations obtained for most of dykes that are close to both the present dipolar field and the directions obtained for the CAMP magmatism from West Africa (Palencia-Ortas et al., 2011) (Fig. 4B). (R6) The Manso pole passes a reversal test (McFadden and McElhinny, 1990) with 5 sites of positive inclination and 8 sites of negative inclination. (R7) The A95 envelope of Manso pole overlaps the robust B2 group ($R = 6$) of Robert et al. (2017) for volcanic units of ~550–530 Ma sampled in the Anti-Atlas mountains, north of the West African Craton. Nevertheless, Meert et al. (2020) underline that comparison to younger poles should be discarded if the younger poles come from an orogenic belt, which is the case with the B2 pole, calculated from units folded during the Carboniferous–Permian Hercynian (or Variscan) orogeny (Robert et al., 2017). Therefore, a remagnetization of the Manso dykes seems improbable because no evidence of deformation has been documented in the study area, and our large spatial sampling discard any localized tectonic rotations. Thus, we can consider that the Manso pole fulfills all the R7 criteria of Meert et al. (2020). The ~860 Ma Manso paleomagnetic pole can thus be considered as a high-quality key pole and can serve as the first Neoproterozoic reference for the paleogeography of West Africa during Rodinia.

5.2. Testing the existence of the long-lived WABAMGO configuration in Rodinia

The Grenville, Sveconorwegian, and Sunsás orogens are typically used to reflect the collision between Laurentia, Baltica and Amazonia between ~1200 and 1000 Ma assembling the Rodinia supercontinent (Hoffman, 1991). This combination between these cratons is nearly identical in several Rodinia reconstructions but was questioned by Meert and Torsvik (2003), and more recently by Evans (2009) using an updated paleomagnetic database. Recently, Martin et al. (2020) re-evaluated the isotopic signature of the long-lived Paleo– to Mesoproterozoic accretionary orogens along the margins of Laurentia, Baltica, Amazonia, and Kalahari. In their model, the core of Rodinia is defined by the Laurentia-Baltica-Amazonia-Kalahari connection finally sutured during the Stenian. In the Rodinian reconstructions, West Africa is still associated with the Amazonia Craton but this link is unconstrained by paleomagnetic data during the Neoproterozoic (D'Agrella-Filho et al., 2016). Using the Manso pole, we can test if a long-lasting connection is paleomagnetically viable between West Africa, Amazonia, and the paleomagnetically well-constrained Baltica and Laurentia block.

A long-lived connection is usually proposed for Baltica and Laurentia between ~1760 Ma and ~ 1270 Ma (Salminen et al., 2014). A large distance is observed across the APW path of Baltica between the mean 1265 Ma pole (BA1) (Pesonen et al., 2003), the Salla diabase VGP (BA2) and the 1100–1050 Ma poles (BA3, BA4) (Table 2; Fig. 7A) (Mertanen et al., 1996; Pesonen et al., 2003) and suggests fast drifting at that time. Fast drifting is also observed for Laurentia with the Logan loop and the ~1105–1080 Ma Keweenawan track (LA1-LA2) (Swanson-Hysell et al., 2019). Between ~1050 and 900 Ma the APW path of Laurentia was defined as the Grenville loop and the same interval in Baltica was defined as the Sveconorwegian loop, but its shape and motion are debated (Elming et al., 2014; Gong et al., 2018). The best attempts to adjust the Grenville and Sveconorwegian loops are along an Euler pole of 75.8°N, 264.2°E, −59.2° (Pisarevsky et al., 2003), or in a tighter fit of 81.5°N, 250°E, −50° (Evans, 2009). However, Baltica rotated ~70° clockwise in relation to Laurentia between the ~1800–1200 Ma NENA (Northern Europe-North America) configuration (Gower et al., 1990) and the ~1050–800 Ma Rodinia configuration (Salminen et al., 2009). These reconstructions imply a single Laurentia-Baltica block between ~1050 and 800 Ma during the Rodinia. Key 951–935 Ma mean poles for Baltica (BA6) confirmed this clockwise motion, but a more complex shape for the APW path is proposed considering the ~971 Ma VGP of Blekinge-Dalarma dykes (Group. B) (BA5) (Fig. 7A) (Gong et al., 2018). A late Svecornorwegian group of poles (936–850 Ma) obtained for the Rogaland Igneous Complex (RIC) and the ~860 Ma Hunnedalen dykes with a stable remanence of same polarity suggest a stable position for Baltica at high latitude (BA7-BA11) (Walderhaug et al., 1999). The primary remanence of this group is supported by an inverse contact test with the ~635 Ma Egersund dyke swarm (Walderhaug et al., 2007).

West Africa is always associated with the Amazonia Craton in the Paleo-Mesoproterozoic reconstructions but in a different position from the Gondwana link (Onstott and Hargraves, 1981). These cratons were juxtaposed with the Baltica in the South America Baltica (SAMBA) connection with the possibility of a long-lived connection between ~1800 and 800 Ma (Johansson, 2009). An alternative reconstruction, using geological considerations, was proposed with Kalahari-Congo, São Francisco and India in the Umkondia supercraton at ~1110 Ma (Choudhary et al., 2019). Using the paleomagnetic poles of the ~1199 Ma Nova Floresta formation (AM1) (D'Agrella-Filho et al., 2008) and ~ 1149 Ma Fortuna formation (AM2) (Tohver et al., 2002) (Fig. 7A), a strike-slip migration of Amazonia along the Laurentia at ~1200–1150 Ma was initially proposed. But, a preliminary ~1110 Ma paleomagnetic pole for the well-dated Rincón del Tigre Complex (AM3) (Teixeira et al., 2015) supports a moderate to low latitude for Amazonia at that time (Patroni, 2015), which is incompatible with the model of oblique collision proposed by Tohver et al. (2002). These AM3 pole for Amazonia supports the model of Evans (2013) suggesting a clockwise rotation of the Amazonia and Baltica to collide with Laurentia. For West Africa, paleomagnetic results were obtained from the Mesoproterozoic units of the Adrar (NW- West Africa, Mauritania) (Morris and Carmichael, 1978; Perrin et al., 1988; Perrin and Prévot, 1988). Rooney et al. (2010) obtained a new *Re*—Os age of ~1105 Ma for the Atar group previously estimated at 890–775 Ma. Based on the similarity with younger directions some authors have suggested some remagnetization effects (Perrin and Prévot, 1988; Tohver et al., 2006), but without further evidence the primary origin for their remanence cannot be ruled out. The characteristic component for those rocks is carried by hematite and was revealed at high temperatures (>590 °C). If the magnetization is primary this means the I9 pole (WA1) calculated by Perrin and Prévot (1988) is coeval to the ~1110 Ma Rincón del Tigre pole (AM3) from the Amazonia Craton (Patroni, 2015) (Fig .7A). No data are available to define the APW path for the Amazonia Craton in Early Neoproterozoic times, but our new ~860 Ma Manso pole satisfies a maximum R-criteria to be considered as an anchor point for paleogeographic reconstructions (Fig. 7A; $R = 7$, this study).

The Mesoproterozoic APW path for the Congo-São Francisco Craton is defined by the well-dated Late Kibaran pole (C1) at ~1236 Ma (Meert et al., 1994b). A large shift is observed between the ~1110 Ma pole from the Huile-Epembe dykes (Salminen et al., 2018) and the ~925 Ma poles from the São Francisco dykes (Salvador, Ilhéus, Oliveira) (Fig. 7A) D'Agrella-Filho et al. (1990); (D'Agrella-Filho et al., 2004; Evans et al., 2016). The Congo-São Francisco Craton is usually associated with the Rodinia supercontinent (Merdith et al., 2017), or in a different view connected with the African blocks (D'Agrella-Filho and Cordani, 2017).

Table 2
Paleomagnetic database for the Rodinia. Code, Plat – pole latitude, Plon – Pole longitude, Age – nominal age, A95 confidence cones of the paleomagnetic poles used in the paleogeographical reconstructions. R-criteria from Meert et al. (2020). ***References:*** **Laurentia**: [1] Park (1981), [2] Swanson-Hysell et al. (2019), **Baltica:** [3] Walderhaug et al. (1999), [4] Brown and McEnroe (2004), [5] Stearn and Piper (1984), [6] Walderhaug et al. (2007), [7] Brown and McEnroe (2015), [8] Gong et al. (2018), [9] Mertanen et al. (1996), [10] Pesonen et al. (2003), [11] Salminen et al. (2009), **West Africa:** [12] **This study**, [13] Perrin and Prévot (1988), [14] Rooney et al. (2010), **Amazonia:** [15] Patroni (2015), [16] Teixeira et al. (2015) [17] D'Agrella-Filho et al. (2008), [18] Tohver et al. (2002), **Congo-São Francisco:** [19] Evans et al. (2016), [20] D'Agrella-Filho et al. (2004), [21] D'Agrella-Filho et al. (1990), [22] Salminen et al. (2018), [23] Meert et al. (1994a), **Kalahari:** [24] Swanson-Hysell et al. (2015a), **North China:** [25] Fu et al. (2015), **Tarim:** [26] Wen et al. (2018).

Code	Name	Age (Ma)	Plat (°N)	Plong (°E)	A_{95}	R1	R2	R3	R4	R5	R6	R7	Rc	References
LAURENTIA														
LA3	Basal units from the Little Dal Group	**892–849**	−15.7	141.3	3.6	0	1	1	1	1	0	0	**4**	**[1]**
LA2	1105 Ma synthetic Keweenawan track	**1105**	42.7	206.4	2.1	–	–	–	–	–	–	–	**>5**	**[2]**
LA1	1110 Ma synthetic Keweenawan track	**1100**	46.3	222.7	3.2	–	–	–	–	–	–	–	**>5**	**[2]**
BALTICA														
BA11	Hunnedalen dykes	**860**	−41	222	10.5	1	1	1	0	1	0	1	**5**	**[3]**
BA10	Egersund-Ogna anorthosite	**870**	−42.1	200.4	9	1	1	1	0	0	0	1	**4**	**[4]**
BA9	Other Rogaland anorthosites	**870**	−43.5	213.7	3.6	1	1	1	0	0	0	1	**4**	**[3, 5]**
BA8	Rogaland Igneous Complex (RIC)	**870**	−45.9	238.4	18.2	1	0	1	1	0	0	1	**4**	**[6]**
BA7	Bjerkreim-Sokndal layered intrusion (RIC)	**916**	−35.9	217.9	6	1	1	1	0	1	0	1	**5**	**[7]**
BA6	Mean 951–935 Ma Baltica pole	**940**	−2.6	239.6	5.8	1	1	1	1	1	1	1	**7**	**[8]**
BA5	971 Ma BDD dykes (VGP)	**971**	−27	230.4	14.9	1	0	1	0	1	1	1	**5**	**[8]**
BA4	Laanila-Ristijärvi dykes	**1050**	−2.1	212.2	16.4	1	0	1	1	1	0	0	**4**	**[9]**
BA3	Mean Baltica 1100 Ma	**1100**	1	208	16	–	–	–	–	–	–	–	–	**[10]**
BA2	Salla diabase dyke VGP	**1122**	71	113	8.1	1	0	1	1	1	0	0	**4**	**[11]**
BA1	Mean Baltica 1265 Ma	**1265**	4	158	4	–	–	–	–	–	–	–	–	**[10]**
WEST AFRICA														
WA2	**Manso dykes**	**860**	**−28.3**	**177.6**	**12.7**	**1**	**1**	**1**	**1**	**1**	**1**	**1**	**7**	**[12] This study**
WA1	Atar group, unit I9 (GH-comp)	**1105**	−41.7	61.2	6.4	1	0	1	0	1	0	0	**3**	**[13, 14]**
AMAZONIA														
AM3	Rincón del Tigre Complex	**1100**	−30	28.5	12.8	1	0	1	0	0	1	1	**4**	**[15, 16]**
AM2	Fortuna formation red beds	**1149**	−59.8	335.9	9	1	1	1	0	1	0	0	**4**	**[17]**
AM1	Nova Floresta formation	**1199**	−24.6	344.6	6.2	1	1	1	0	1	0	0	**4**	**[18]**
CONGO-SAO FRANCISCO														
São Francisco														
SF3	Bahia Coastal dykes	**924**	−7.3	286.4	6.2	1	1	1	1	1	0	1	**6**	**[19]**
SF2	Salvador dykes -N	**924**	−6.4	302.7	15.6	1	1	1	1	1	0	0	**5**	**[20]**
SF1	Itaju do Colônia	**924**	−7.7	291	9.8	1	1	0	1	1	0	0	**4**	**[21]**
Congo														
C2	Huila-Epembe dykes	**1105**	−34.7	256.5	8.7	1	1	1	1	1	1	1	**7**	**[22]**
C1	Late Kibaran intrusives	**1236**	17	292.7	5.9	1	1	1	0	0	0	0	**3**	**[23]**
KALAHARI														
K1	Umkomdo grand mean pole	**1110**	−64	222.1	2.6	1	1	1	1	1	1	1	**7**	**[24]**
NORTH CHINA														
NC1	Wangshan sills of Huabei -C	**895**	52.3	329.3	3.5	1	1	1	1	1	1	0	**6**	**[25]**
TARIM														
T1	Salajiazitage Group volcanics	**880**	25.5	217	11.3	1	1	1	1	1	0	0	**5**	**[26]**

Similarities in length and shape of the APW paths for Baltica, Amazonia, West Africa, and São Francisco-Congo cratons allow us to build a Master APW path between ~1200 and 850 Ma for these continental units (Fig. 7A). This Master APW path suggests that these blocks were nearby between ~1200 and 850 Ma, forming a single continental entity. This large continental unit, hereafter referred to WABAMGO (West Africa-Baltica-Amazonia-Congo) is represented in Fig. 7B. Geological domains, LIP comparisons, and paleomagnetic data between Baltica, Amazonia, and West Africa support a strong connection between these continents since the Paleoproterozoic using the SAMBA model (Baratoux et al., 2019; D'Agrella-Filho et al., 2016; Terentiev and Santosh, 2020). Our model supports the presence of a long-lived accretionary margin during the Mesoproterozoic times in the western part of Amazonia (Sadowski and Bettencourt, 1996). The orientation of West Africa in relation to Amazonia in the SAMBA model is paleomagnetically viable from ~2100 Ma until ~1400 Ma (D'Agrella-Filho et al., 2016), but a later reorientation is necessary to fit with the classical ~550 Ma Gondwana configuration. In the WABAMGO configuration, the West Africa-Amazonia connection is close to this late Neoproterozoic configuration, suggesting this reorientation could have occurred between ~1400 and 1200 Ma, during the breakup of the Columbia supercontinent. The Sunsás and Sveconorwegian orogens are correlated between Amazonia and Baltica in this model (Fig. 7B). A voluminous Early Tonian mafic magmatism is observed in the São Francisco Craton spanning from ~925 Ma to 850 Ma (Danderfer et al., 2009; Moreira et al., 2020). Similar records of Tonian age magmatism have been reported for the Congo with the Sembé Ouesso (Vicat and Pouclet, 1995), for Baltica with the ~850 Ma Hunnedalen dykes (Walderhaug et al., 1999), for West Africa with the ~860 Manso and ~ 915 Oda dykes in the south of the craton (Baratoux et al., 2019) and the ~880–850 Iguerda-Taïfast dykes in the Anti-Atlas (Kouyaté et al., 2013). A potential match in Amazonia with these ~870–850 Ma LIPs could be the Makinet dykes (formerly Tampok) that were dated with a K—Ar cooling age of 809 ± 58 Ma (at 2-sigma level) (Delor et al., 2003). Early Tonian rifting is also documented for the northern part of the West African Craton with the volcanoclastic deposits of the ~883 Ma Tachdamt Fm (Bouougri et al., 2020). Coeval Early Tonian extension in West Africa and São Francisco is thus supported by the WABAMGO configuration (Fig. 7B). A rare "Grenvillian event" (~1000–920 Ma) documented within the WABAMGO juxtaposition is the Cariris Velhos tectonic event, proving some continental

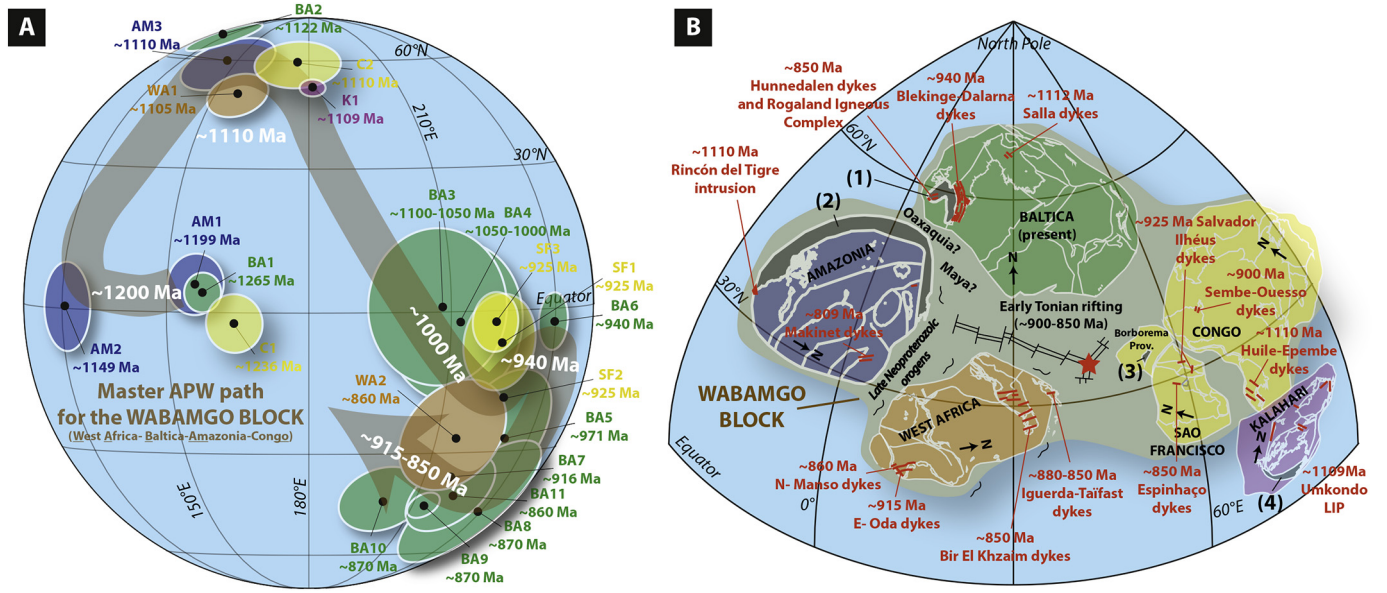


Fig. 7. A: The WABAMGO (West Africa-Baltica-Amazonia-Congo) juxtaposition using Baltica as reference frame in present coordinates. Euler poles used in the Baltica reference frame for the different cratons in the WABAMGO configuration: West Africa (1.375°N, 15.1585°E, −66.0819°), Amazonia (10.894°S, 369.8179°E, −89.4471°), Congo (52.32°S, 158.71°E, −70.57°). The São Francisco Craton is rotated to Congo using its pre-Mesozoic configuration (46.8°N, 329.4°E, 55°, (McElhinny et al., 2003)). Kalahari is connected to the Congo Craton (12.8°S, 19°E, −43.2°) according the configuration of Salminen et al. (2018). Present North is indicated for each craton by an arrow. *Abbreviations for the ~ 1200–1000 Ma Grenvillian orogens (in black):* (1) Sveconorwegian orogen, (2) Sunsás orogen, (3) Cariris Velhos event, (4) Namaqua-Natal orogen. Tonian large igneous provinces (LIPs, in red) are indicated. B: Master apparent polar wander (APW) path for the WABAMGO between ~1200 and 850 Ma. Paleomagnetic poles and used abbreviations are listed in the Table 2. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

reorganization and convergence at the edge of the mega block of the Borborema Province was followed by a Tonian rifting (dos Santos et al., 2010; Neves et al., 2020).

Detrital zircon ages on Neoproterozoic units from northern and southern West Africa (Ait Lahna et al., 2020; Bouougri et al., 2020; Kalsbeek et al., 2008) indicate "Grenvillian" ages (~1250–925 Ma). The potential sources of these Mesoproterozoic-Neoproterozoic zircon grains could be explained by the proximity of the Grenvillian orogens to the west such as Sunsás and Sveconorwegian and the associated small blocks (*e.g.* Oaxaquia, Maya), and the Cariri Velhos event to the east. Major shifts in the sedimentation on the West African Craton suggest a complete breakup of the WABAMGO from Baltica and São Francisco-Congo after ~800 Ma with the development of subduction zones in northern West Africa in the late Neoproterozoic (Triantafyllou et al., 2016). Following this model, the proximity of the Congo-São Francisco and West Africa may explain the large displacements along the major dextral Transbrasiliano lineament with the closure of the Pharusian Ocean during the assembly of Gondwana in late Neoproterozoic times. The proposed long-lived WABAMGO juxtaposition is geologically and paleomagnetically viable between ~1200 and 800 Ma.

5.3. Implications for the Rodinia assembly

Using our new Master APW path for the West Africa-Baltica-Amazonia-Congo block, the reconstruction at ~1110 Ma shows that the assembly of Rodinia is dominated by the V-shape closure of the external Nuna Ocean (Li et al., 2019), or Grenville Ocean, (Sadowski and Bettencourt, 1996), between the northern Laurentian blocks and the southern WABAMGO (Fig. 8A). This hypothesis of the closure of this Ocean by accretionary orogens was previously suggested on geological grounds (Cawood and Pisarevsky, 2017; Martin et al., 2020), but our study add a paleomagnetic support. The Kalahari Craton is positioned in Fig. 8 along the southern tip of Congo (pos-A) as in (Salminen et al., 2018) or in its inverted position (pos—B) with the Natal-Namaqua orogen facing the coeval Grenville orogen in Laurentia, in both cases constrained by the ~1109 Ma Umkondo pole (K1) (Swanson-Hysell et al., 2015a). In the preferred position B, the Kalahari Craton will collide with the southern coast of Laurentia after the closure of the external Nuna Ocean (Fig. 8B). Nevertheless, the Kalahari position is poorly constrained in Rodinia, and is beyond the scope of this contribution. The orientation of West Africa in the WABAMGO configuration (pos-A, Fig. 8B) differs from the SAMBA model, in which the western margin of West Africa is associated to the southern part of Baltica. Nevertheless, our new proposed configuration requires new Tonian high-quality poles for the unconstrained cratons, especially Amazonia, and new Mesoproterozoic poles for West Africa to confirm the relative orientation between Amazonia and West Africa. Discarding the ~1105 Ma WA1 pole, a position closer to the SAMBA model (Johansson, 2009) is paleomagnetically plausible for West Africa (pos-B, Fig. 8B).

For the final paleogeographic configuration at 925–850 Ma we can compare the position of West Africa derived from the Manso paleomagnetic pole with the available record of Laurentia and surrounding blocks (Fig. 8C). Three paleomagnetic poles are available for the northern blocks of Rodinia. Laurentia can be constrained by the basal units from the Little Dal Group, LA3 pole, which passes a fold test, and its age was recently constrained by high-quality correlations between 892 and 849 Ma (Greenman et al., 2020; van Acken et al., 2013). One pole for the Tarim (T1) at ~880 Ma (Wen et al., 2018), and one pole for North China at ~895 Ma (NC1) (Fu et al., 2015) complete the paleomagnetic database for the northern blocks of Rodinia. Our model implies that Rodinia was formed by extroversion (Murphy and Nance, 2003), with the closure of an external Nuna Ocean between two large blocks, the southern WABAMGO and the northern Laurentian block. Rodinia was finally assembled by a large dextral motion of the Australia-Tarim blocks as suggested by Wen et al. (2018). The polar location of the WABAMGO at ~900 Ma, together with the low-mid latitudinal distribution of Laurentia and the blocks of North China, Tarim, Australia (?) confirms a pan-latitudinal rather than an equatorially distributed supercontinent as suggested by Jing et al. (2019).

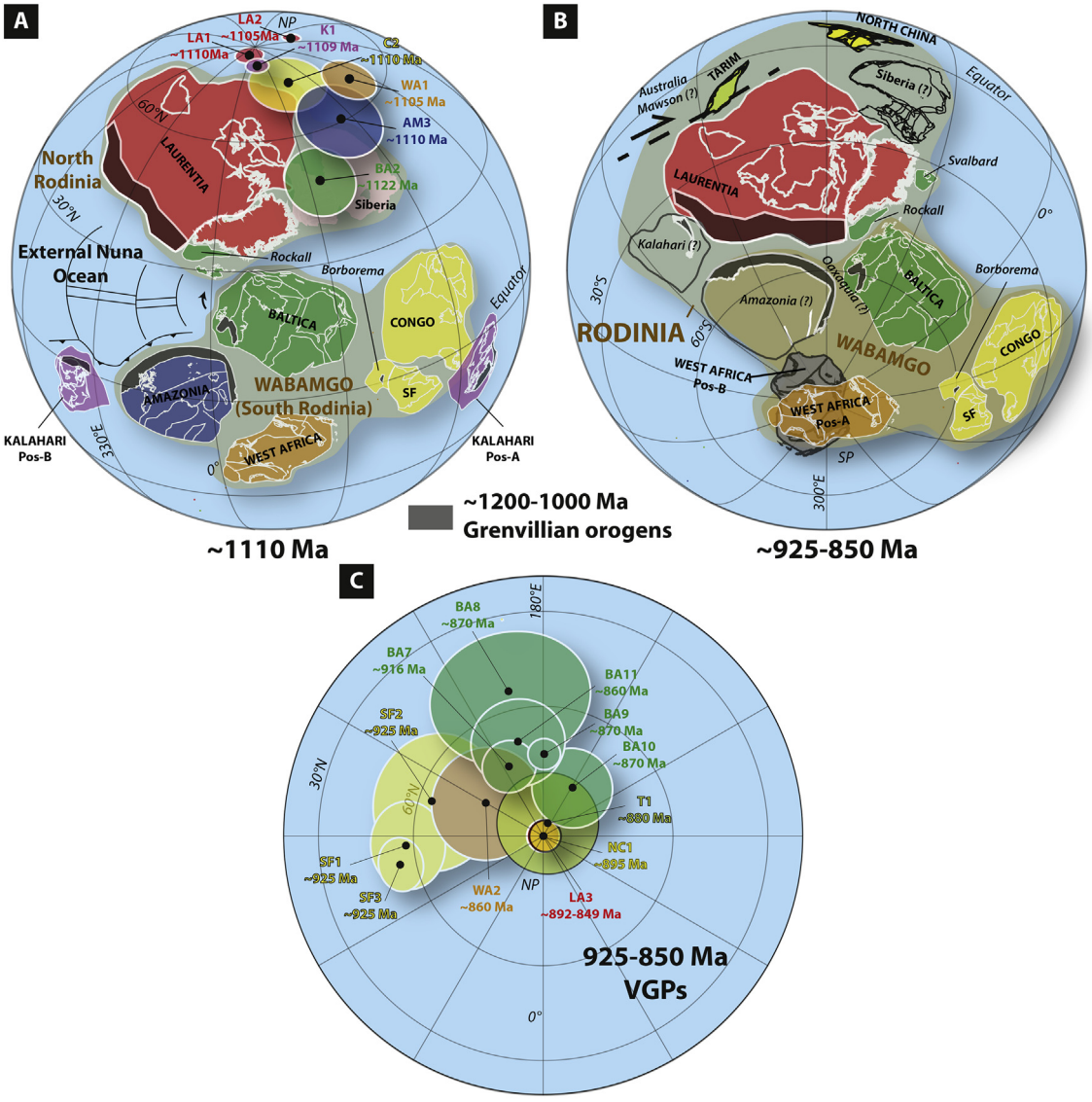


Fig. 8. A: Paleogeography reconstruction of the Rodinia assembly at ~1110 Ma showing the presence of the external Nuna Ocean (Grenville Ocean) between the WABAMGO (south Rodinia) and the Laurentia (north Rodinia). Two plausible positions are illustrated for the Kalahari Craton with the Salminen et al. (2018) configuration (pos-A), and Kalahari placed in the west of Amazonia (pos-B) as paleolongitude is arbitrary. We followed the model of Sadowski and Bettencourt (1996) using a subduction toward Amazonia. B: Paleogeography reconstruction of the Rodinia at ~925–850 Ma. Cratons with paleomagnetic constraints (in colors) and cratons with no data (?) are indicated. Two positions of West Africa are illustrated with the position A in agreement with the WABAMGO configuration and the plausible position B according to the uncertainty for the paleolongitude. The hypothesis of a dextral movements from Wen et al. (2018) is indicated for the northern cratons of Australia-Mawson-Tarim. C: Paleomagnetic poles (north pole) used in the reconstruction at ~925–850 Ma (B). See Table 2 for abbreviations and references.

6. Conclusions

New U—Pb apatite ages of 857.2 ± 8.5 Ma and 855 ± 16 Ma agree with the previous 867 ± 16 Ma U-Pb baddeleyite age (Baratoux et al., 2019) confirming the extension of the Manso dyke swarm to the north of Ghana. A ~ 860 Ma Manso key pole (28.3°S, 177.6°E, A95 = 12.7°, $R = 7$) was calculated and represent the first high-quality paleomagnetic data in Early Tonian for the West African Craton. The remanent magnetization is considered as primary, passing a baked contact test and a reversal test. The paleomagnetic database for West Africa, Baltica, Amazonia, and the Congo-São Francisco cratons supports a long-lived continental unit between ~1200 and 800 Ma, the WABAMGO juxtaposition. During the late Mesoproterozoic-Early Neoproterozoic (~1200–900 Ma), the WABAMGO and the Laurentia were drifting southward. During this migration, a clockwise rotation of the WABAMGO in relation to Laurentia closed the external Nuna Ocean causing the Grenvillian collisions suturing the Rodinia supercontinent. Thus, this model favors with paleomagnetic support the formation of the Rodinia supercontinent by extroversion.

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Sample credit author statement

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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